Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

"To eliminate concern of audio not working in Scorpio and Marazzo model.

Mahindra and Mahindra Ltd, Nasik Plant, Multimodel Assembly Line.

	Plan	Actual
Start Date	**01.06.19**	**01.06.19**
Completion Date	**10.12.19**	**25.12.19**

Current Status: **Closed**

Cross Functional Team

No	Description	Name	Designation	Dept	Phone
1	**Team Leader**	Dnyaneshwar Wagh	L8-O	PRD	9730015208
2	**Mentor**	Sandip Chikhale	L5-M	STCF	9922408138
3	**Member 1**	Sayali More	L7-0	Process	7262090356
4	**Member 2**	Ganesh Bendale	L7-O	PRD	9850952219
5	**Member 3**	Satish Pagar	L8-0	MQA	9822209653
6	**Member 4**	DD Sadavarte	DR	PRD	8007337681
7	**Member 5**	Sadasivam Gobeeswaran	L7-0	SQA	8408004433

Step1. Select Subject

Brief description of Problem:-

To eliminate concern audio not working in Scorpio and Marazzo model.

What : Audio not working

Where : At FAI/CAI

When : Since Jun-19 month

Who : By stage associate,Buy off electric check man at FAI

WHY : High rework, FRC /FBO loss, Repeat issue at Buyoff/FAI/CAI .

HOW : Avg. 15 cases/month reported at FAI and 1-2 cases at CAI

Audio not working & Infotainment function malfunction

2. Justify the choice (reasons)

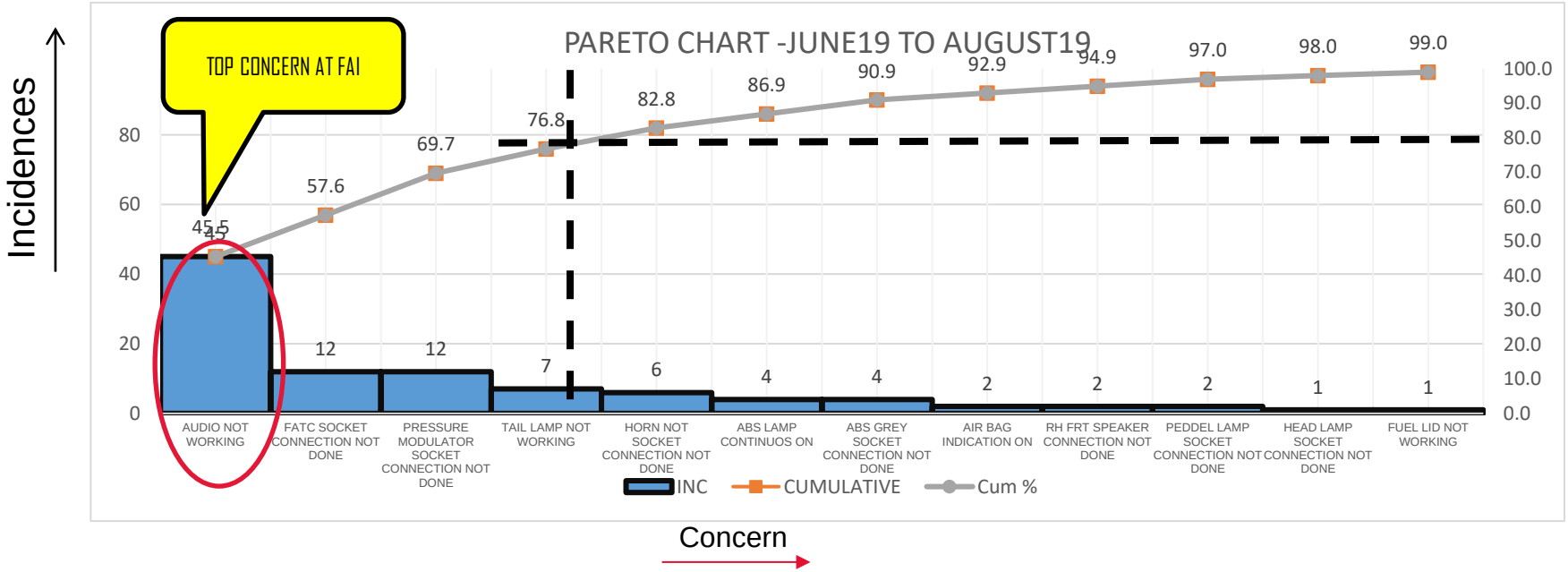
Crusade 2.0

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

2. Justify the choice

PARETO ANALYSIS

TRIM 3 LINE CONCERNS REPORTED AT FAI



Inference:
Audio not working is top concern and repeated chronic concern, hence taken for resolution.

2. Justify the choice

Severity & Consequences of Concern :-

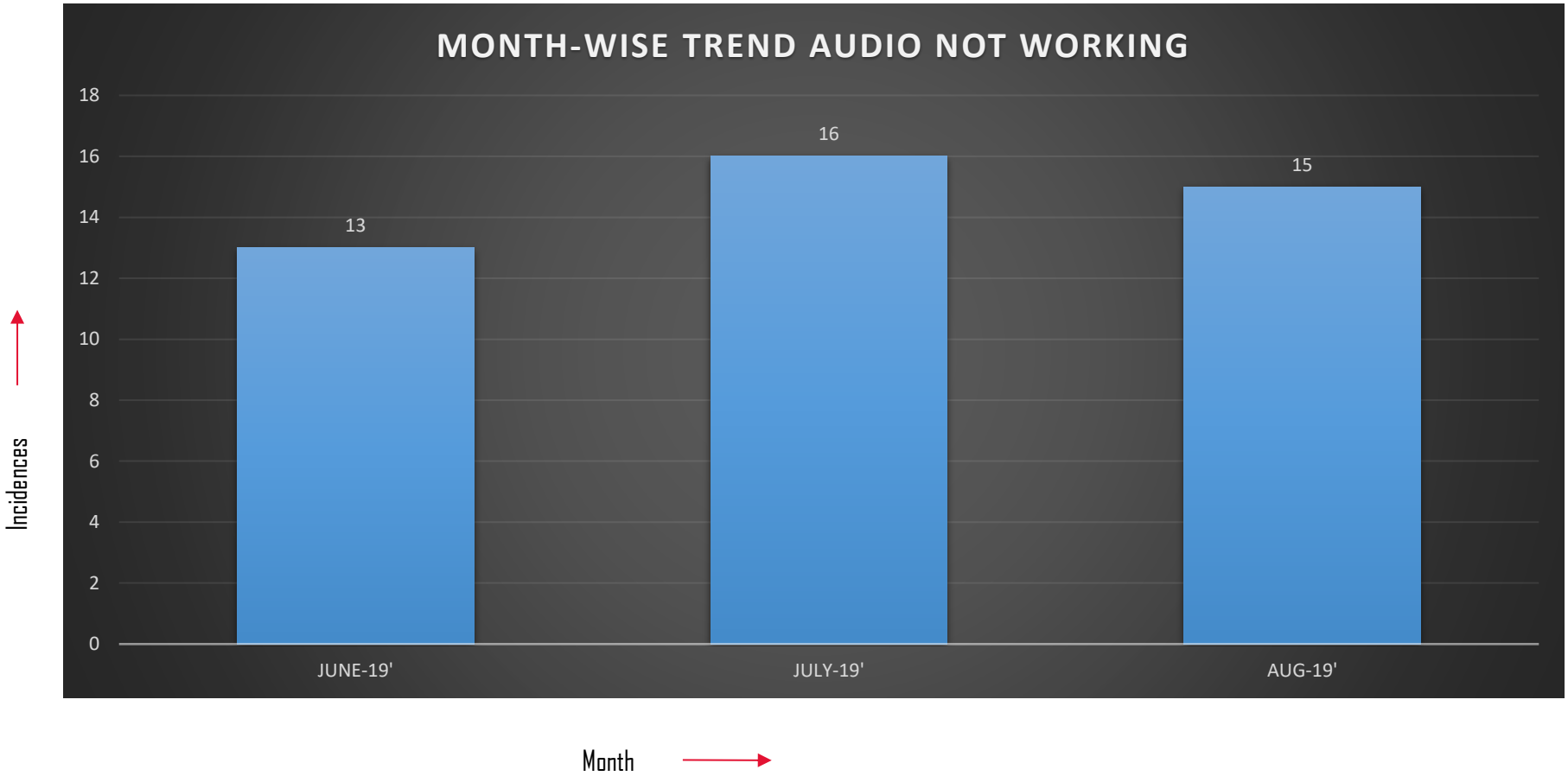
- **Avg. 15 Cases reported per month, for rework, 1-2 concern reported in FAI/CAI daily.**
- **Customer Dissatisfaction due to Audio malfunctioning**
- **Rework Difficult**
- **Rework cost more.**
- **Rework time 30 min (approx.) for each vehicle by 1 operator.**
- **Rework cost is 19103.2 Rs/year**

Cost of Poor Quality - Rework

Repair /Restoration Cost estimation		
Description	Units	
Rework time (Approx)	Min	30
Avg Concerns/Day	Nos	0.5
Total Required Rework time per day	Min	15
Effective working time	Min	465
Operator required /day	nos	0.03
Approx Operator cost/ day	Rs	1974
Operator cost /concern	Rs	63.68
Avg no. of working days	days	25
Operator cost /month	Rs	1591.9
Operator Cost/year	**Rs**	**19103.2**

2. Justify the choice

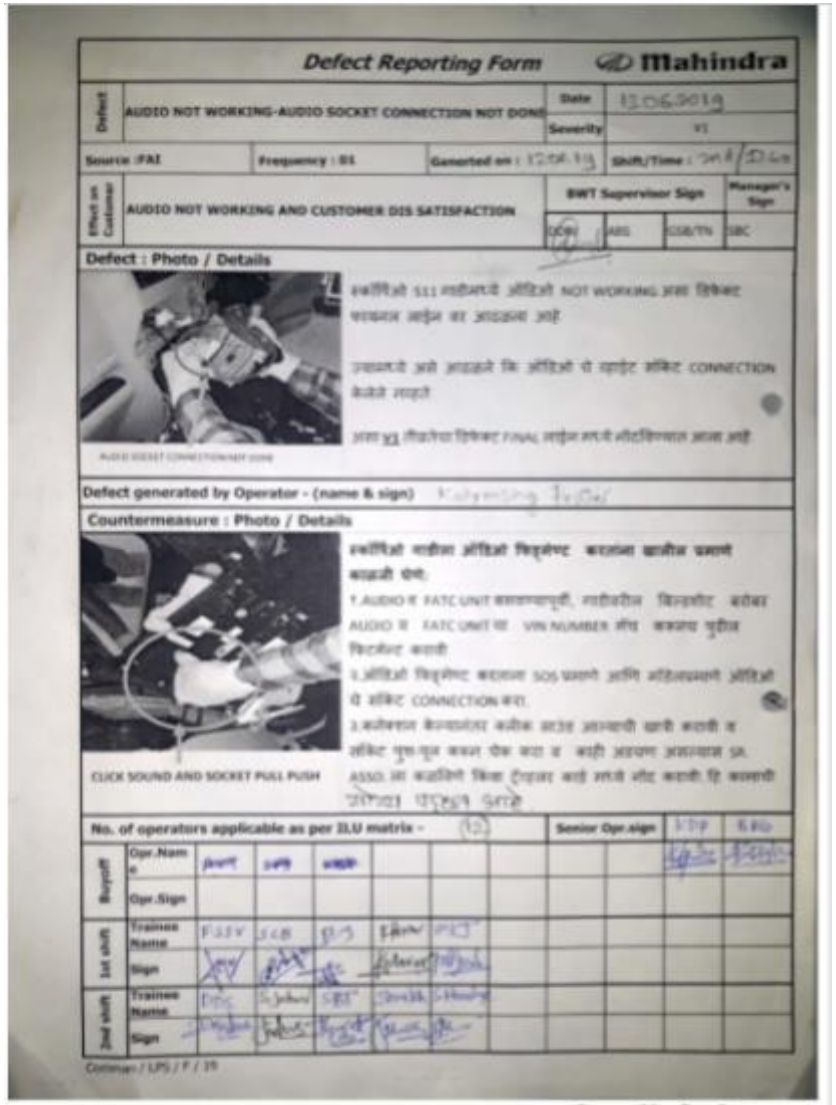
MONTHWISE DEFECT TREAD :-



Inference : Average 15 No.s concerns reported per month from June-19 To Aug-19

2.b Immediate Containment Action

- Training for stage associates, explaining main step and key points.
- Emphasis on Audio Socket connection and fitment as shown.
- 100% lot checked for supplier variability.
- DRF given to stage associates, explaining key points and focus Audio socket connection



Defect Reporting Form — Mahindra

Defect: AUDIO NOT WORKING-AUDIO SOCKET CONNECTION NOT DONE

Date: 13.06.2019

Severity: VI

Source: FAI | Frequency: 01 | Generated on: 13.06.19 | Shift/Time:

Effect on Customer: AUDIO NOT WORKING AND CUSTOMER DIS SATISFACTION

EWT Supervisor Sign | Manager's Sign

Defect : Photo / Details

Defect generated by Operator - (name & sign)

Countermeasure : Photo / Details

CLICK SOUND AND SOCKET PULL PUSH

No. of operators applicable as per ILU matrix -

Senior Opr.sign

Buyoff | Opr.Name | Opr.Sign

1st shift | Trainee Name | Sign

2nd shift | Trainee Name | Sign

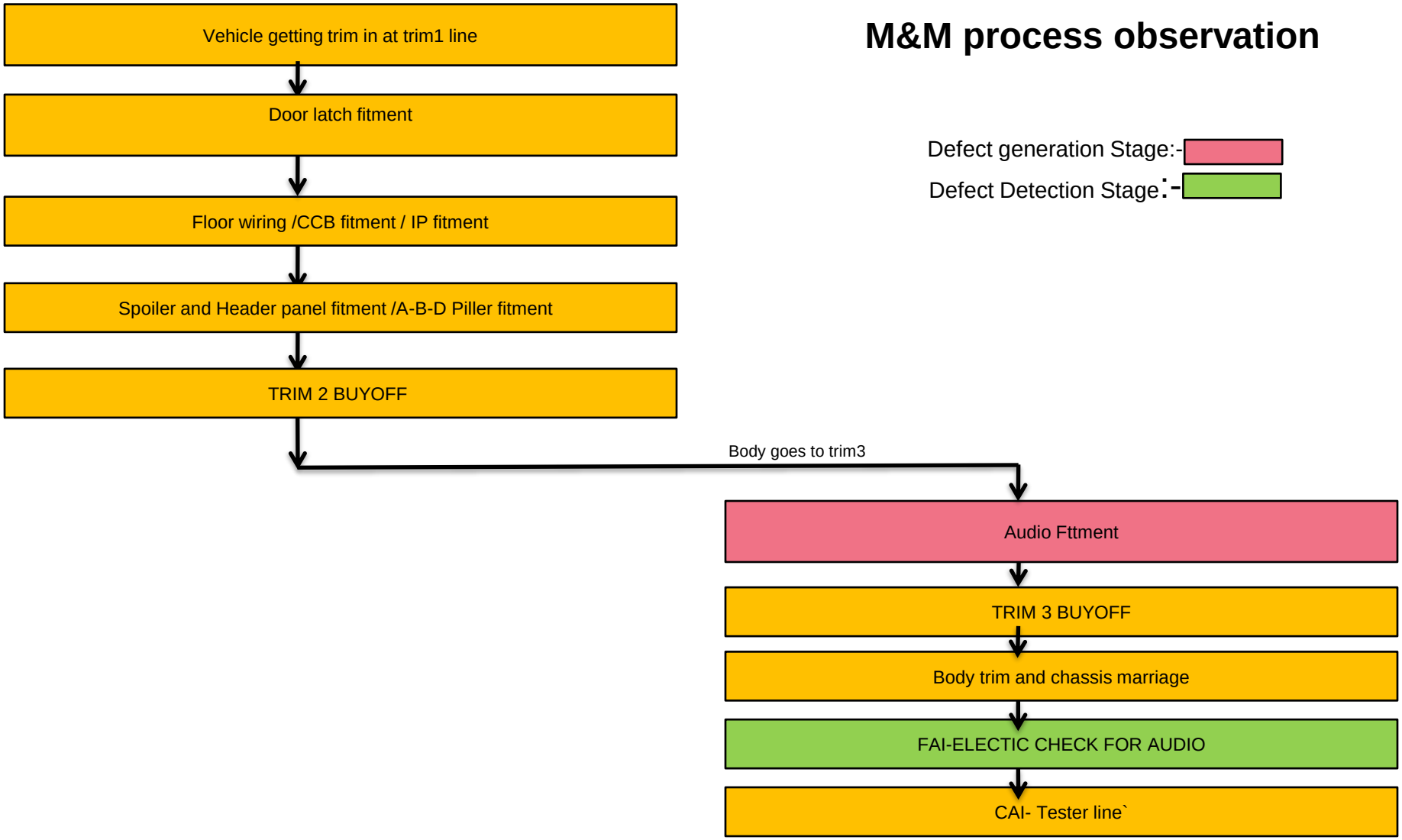
3. Understand the current situation

Flow Chart :- Observation of Process

M&M process observation

Defect generation Stage:-

Defect Detection Stage:-



Step 3. Understanding Current Situation

Observation of process

Pick up Audio and FATC from trolly → **Audio and FATC flashing** → **Ensure success sticker after audio flashing**

↓

Audio socket connection

↓

Audio and FATC fitment on vehicle

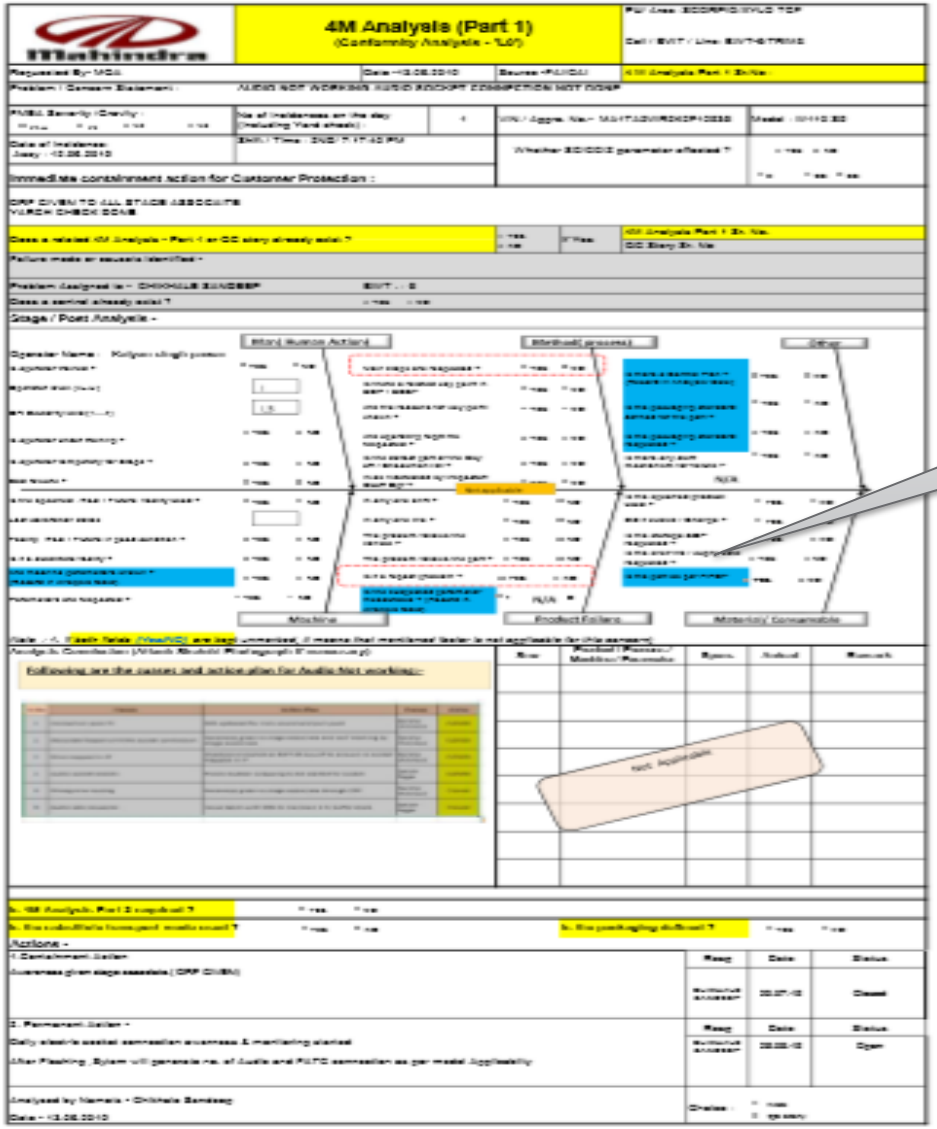
↓

Audio socket connection.

CHECKING AT TRIM3 BUYOFF → **CHECKING AT FAI**

Defect generation stage
Defect identification stage

Step 3. Understanding Current Situation



L0 ANALYSIS FOR AUDIO NOT WORKING

Inference from L0
Associate forgot do do the socket connection
Problem is a repeat problem

4M analysis Good and Not so good observations	
MAN	Operator dext. Level L3
Method	Operator aware of key point and check present at buyoff
Others	Control plan adhered on stage
Machine	Tools used are calibrated
Product failure	Associate forgot to do socket connection
Material	Handling and shelf life of part adhered

V1

NO QUICK WIN

Step 3. Understanding Current Situation

Good & Not so good observations

- Issue observed in Scorpio,Marazzo & W109 vehicle also.
- Issue reported in both the shifts
- Repair time is high, around 12 minutes per vehicle
- Level 3 operators are deployed for Audio fitment.
- Associates are aware of key point & check point at respective stages.
- Audio not working observed in Scorpio and Marazzo models

4. Select targets

Present Level:	Avg. 15 Incidences are being reported at FAI and 1-2 cases reported at CAI
Target Level:	To eliminate the concern of audio not working in Scorpio /Marazzo Model

Step 5. Analysis

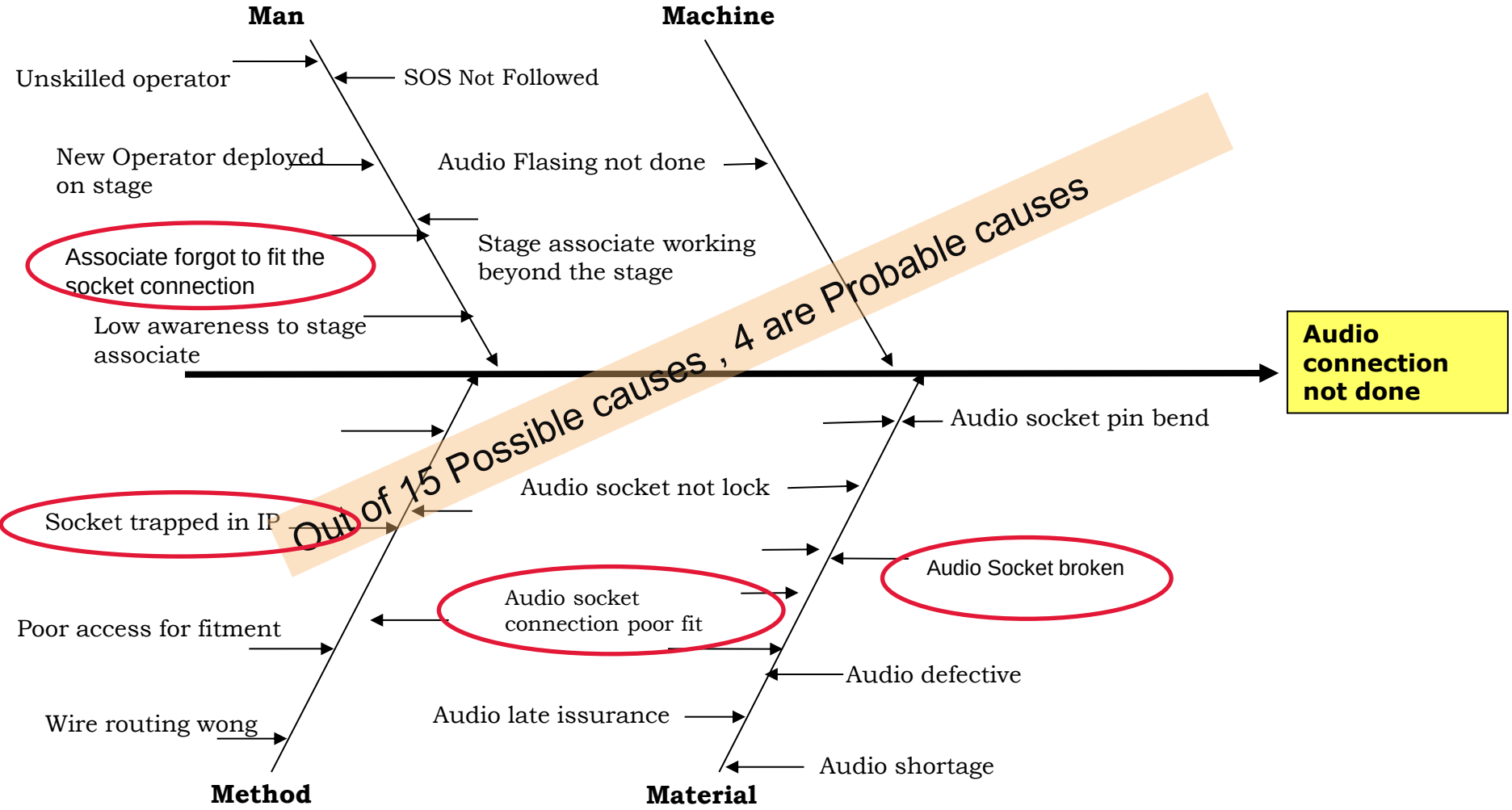
List of Possible causes identified in Brain storming session

Brainstorming Done along with CFT & we have found out 15 Possible Causes for this defect.

S N	Probable Cause
1	Unskilled operator
2	New Operator deployed on stage
3	SOS Not Followed
4	Low awareness to stage assocaite
5	Stage assocaite working beyond stage
6	Associate forgot to fit the socket connection
7	Poor access for fitment
8	Audio socket broken
9	Audio Late issuance
10	Audio socket pin bend
11	Audio Flashing not done.- pin damage/network issue
12	Audio socket not lock
13	Wire trapped in IP
14	Audio shortage
15	Connection poor fit
16	Audio defective
17	Audio socket pin backout

Step 5. Analysis

5.1 Identify possible causes: Fish Bone diagram



Step 5. Analysis

List of Probable causes :

S N	Probable Cause	Ganesh	Wagh	Santosh	Satish	Nandy	SS Shelke	Total
1	Unskilled operator	3	1	3	0	1	1	9
2	New Operator deployed on stage	3	3	3	3	3	1	16
3	SOS Not Followed	0	1	0	0	1	3	5
4	Low awareness to stage assocaite	1	0	3	1	3	0	8
5	Stage assocaite working beyond stage	0	0	0	1	0	1	2
6	Associate forgot to fit the socket connection	9	3	9	3	3	9	36
7	Poor access for fitment	0	0	0	1	0	1	2
8	Audio socket broken	9	3	3	9	3	9	36
9	Audio Late issuance	1	3	0	1	1	0	6
10	Audio socket pin bend	0	3	1	3	0	3	10
11	Audio Flashing not done.- pin damage/network issue	3	3	0	3	3	3	15
12	Audio socket not lock	3	3	3	1	1	3	14
13	Wire trapped in IP	9	3	3	9	3	9	36
14	Audio shortage	0	1	0	3	1	3	8
15	Connection poor fit	9	9	9	9	9	9	54
16	Audio defective	0	1	0	3	1	3	8
17	Audio socket pin backout	3	1	0	3	1	3	11

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting probable cause is 36.

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering.

S N	Not Considered cause	Reason
1	Unskilled operator	All Trained assocaites
2	New Operator deployed on stage	All Trained assocaites
3	SOS Not Followed	25 cycle checked found ok
4	Low awareness to stage assocaite	25 cycle checked found ok
5	Stage assocaite working beyond stage	checked -Found ok
6	Poor access for fitment	30 veh checked found ok
7	Audio Late issuance	Late issuance froom SCM
8	Audio socket pin bend	Visually checked_Found ok
9	Audio Flashing not done.- pin damage/network issue	Visually checked_Found ok
10	Audio socket not lock	Click sound of socket
11	Audio shortage	Late issuance froom SCM
12	Audio defective	Found ok
13	Audio socket pin backout	Found ok

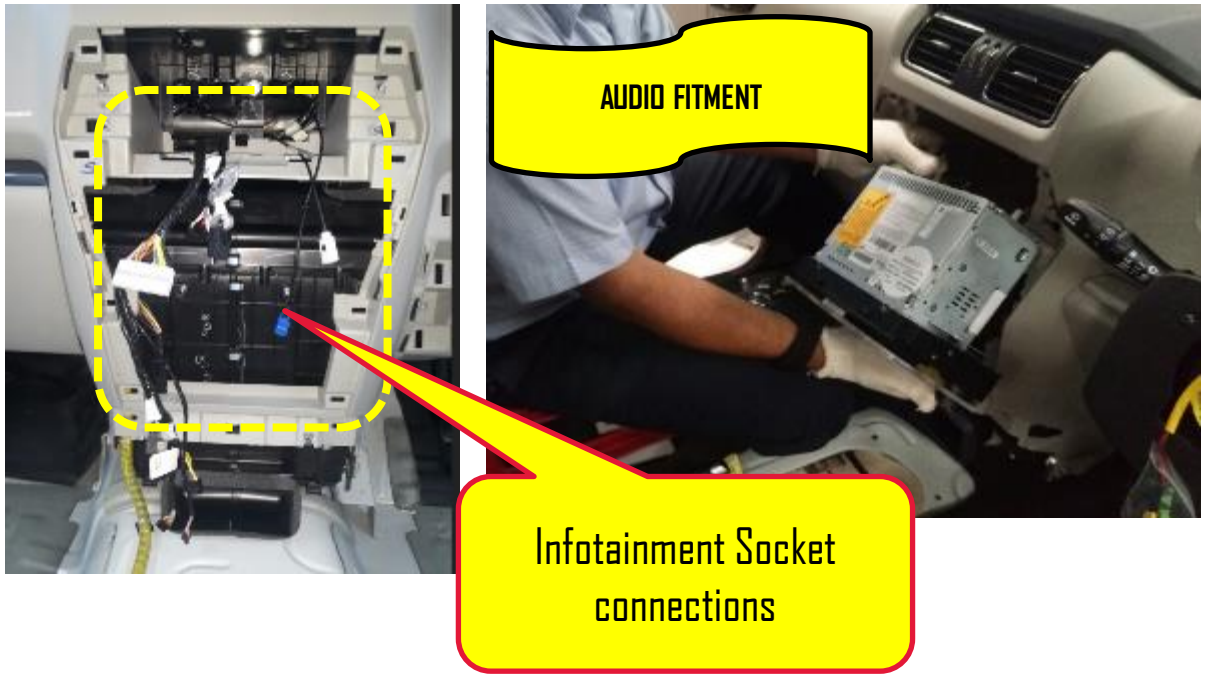
5.1 Identify Probable cause

Sr.No	Probable Cause
1	Associate forgot to fit the socket connection
2	Audio socket broken
3	Wire trapped in IP
4	Connection poor fit

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Associate forgot to fit the socket connection	Visual Check	It was found that,as per model applicability,socket connections are different for different variants. Hence stage associate forgot to do one of audio socket connection.	**Hypothesis Valid**



ANALYSIS AND CONCLUSION :

1. As shown,it was found that, Socket connection are different for different variants.
2. As socket connections are different for different variants,stage associate is not aware of socket connection applicability.
3. Stage associate may forget to do connection due to different variants

AUDIO SOCKET CONNECTION MATRIX

Sr .No	MODEL	AUDIO (No.of Connections)	FATC/CETRAL BEZEL (No.of Connections)
1	W110-S11	5	5
2	S7	4	2
	S5	NA	1
3	M8	6	2
4	M6	5	3
5	M4	4	3
6	M2	NA	3
7	W109 HIGH	4	5
8	W109 MID	4	2
9	W109 LOW	NA	1
10	W110S9	5	5

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Wire trapped in IP	Visual Check	Total=50veh checked.It was found that, 7veh not ok with wire trapped in IP. As socket is trapped,stage associate not able to do connection and hence defect report as audio not working.	**Hypothesis Valid**

DATA ANALYSIS FOR AUDIO WIRE TRAPPED

srNo	VIN No.	Model Description	Remarks
1	MA1RB2VG2K2F13537	W109 SC LHD mHawk EV 2WD - ESP & ABAG MID - AW	Wire trapped
2	MA1RD4VH2K2F15772	W109-DC LHD mHawk EVI 4WD AIRBAG & ESC-HIGH - NB	Wire trapped
3	MA1RB2VH2L2016796	W109 - SC LHD mHawk EVI 2WD AIRBAG & ESC - MID	Wire trapped
4	MA1TA2WR2K2F12423	W110 S11 2WD 8FF ABS A/BAG 140HP	ok
5	MA1TA4WR2K2F12482	W110 S11 4WD 7SF ABS A/BAG 140HP	ok
6	MA1TA4WR2K2F12491	W110 S11 4WD 7SF ABS A/BAG 140HP	ok
7	MA1TA2WR2K2F12856	W110 S11 2WD 8FF ABS A/BAG 140HP	ok
8	MA1RA4VK2L2017971	SKD-W109 SC RHD EII 4WD ABAG MID- w/o Doors & Bnt	Wire trapped
9	MA1RB4VH2L2018007	W109 - SC LHD mHawk EVI 4WD AIRBAG & ESC - HIGH	Wire trapped
10	MA1TA4WR2K2F12500	W110 S11 4WD 7SF ABS A/BAG 140HP	ok
11	MA1TA2WR2K2F12561	W110 S11 2WD 8FF ABS A/BAG 140HP	ok
12	MA1RD4VK2L2018218	W109 DC LHD mHawk EII 4WD ABS ABAG MID-DSAT	Wire trapped
13	MA1TA2WR2K2G18780	W110 S9 2WD 7SF ABS A/BAG 140HP	Wire trapped
14	MA1RD4VH2L2019710	W109 - DC LHD mHawk EVI 4WD AIRBAG & ESP - MID	Wire trapped
15	MA1TA2WR2K2H22987	W110 S11 2WD 7SF ABS A/BAG 140HP	ok
16	MA1RA4VK2L2023563	W109 SC RHD mHawk EII 4WD ABS A.BAG MID-AW	ok
17	MA1TA2WR2K2H23685	W110 S11 2WD 7SF ABS A/BAG 140HP	ok
18	MA1TA2WR2K2H24234	W110 S7 2WD 7SF ABS A/BAG 140HP	ok
19	MA1RA4VK2L2024614	W109 SC RHD mHawk EII 4WD ABS A.BAG MID-AW	ok
20	MA1TA2WR2K2F12356	W110 S11 2WD 8FF ABS A/BAG 140HP	ok
21	MA1TA2WR2K2F12641	W110 S11 2WD 8FF ABS A/BAG 140HP	ok
22	MA1TA4WR2K2F12643	W110 S11 4WD 7SF ABS A/BAG 140HP	ok
23	MA1TA2WR2K2F12644	W110 S11 2WD 8FF ABS A/BAG 140HP	ok
24	MA1TA4WR2K2F12655	W110 S11 4WD 7SF ABS A/BAG 140HP	ok
25	MA1TA4WR2K2F12667	W110 S11 4WD 7SF ABS A/BAG 140HP	ok
26	MA1TA4WR2K2F12671	W110 S11 4WD 7SF ABS A/BAG 140HP	ok
27	MA1TA2WR2K2F12723	W110 S11 2WD 8FF ABS A/BAG 140HP	ok
28	MA1TA4WR2K2F12764	W110 S11 4WD 7SF ABS A/BAG 140HP	ok
29	MA1TA2WR2K2F12396	W110 S11 2WD 8FF ABS A/BAG 140HP	ok
30	MA1TA2WR2K2F12860	W110 S11 2WD 8FF ABS A/BAG 140HP	ok
31	MA1TA2WR2K2F12873	W110 S11 2WD 8FF ABS A/BAG 140HP	ok
32	MA1TA2WR2K2F12887	W110 S11 2WD 8FF ABS A/BAG 140HP	ok

As per data, wire trapped is repeated issue.

Conclusion :–

It was found that, as wire is trapped under IP,stage associate unable to do Infotainment socket connection

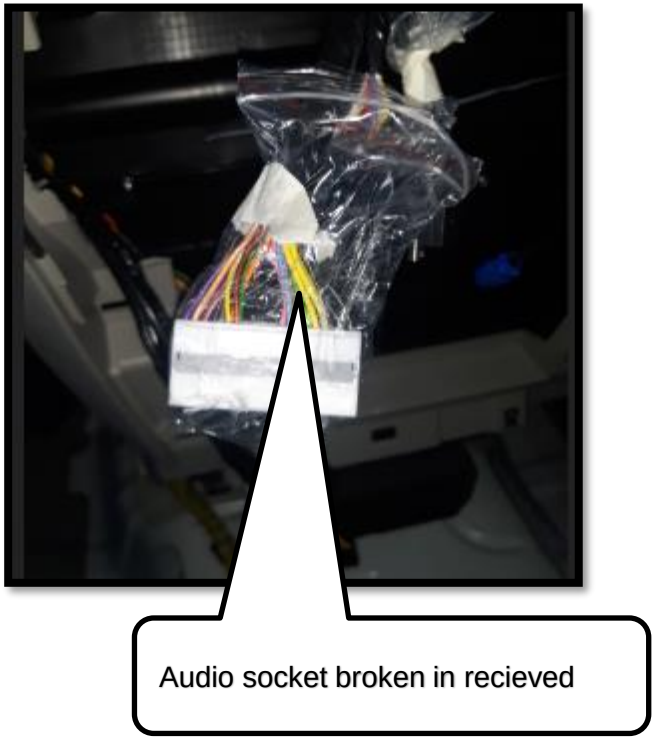
Hence Hypothesis is valid

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	Audio socket broken	Visual Check	It was found that socket get broken due to poor or multiple part handling handling	Hypothesis valid

Audio Socket Broken



Conclusion:

1.It is observed that, Audio socket is getting broken in recieved condition.

2. This is due multiple handling of wiring harness.

3. Unknowingly Socket gets broken while part handling.

4. As socket is broken,stage associate unable to connect the socket.

5.Hence Hypothesis is Valid

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	Connection poor fit	Visual check	It is found that, stage associate not following the SOS_keypoint.Hence socket connection poor fit.	Hypothesis valid

AUDIO SOCKET CONNECTION

CONCLUSION:
Stage associate is not following the SOS- Keypoint. This result into poor locking of sockets. Hence Hypothesis is valid

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Conclusion
1	Associate forgot to fit the socket connection	Visual Check	**Hypothesis valid**
2	Wire trapped in IP	Visual Check	**Hypothesis valid**
3	Audio socket broken	Visual Check	**Hypothesis valid**
4	Connection poor fit	Visual Check	**Hypothesis valid**

4-Valid Causes

WHY - WHY Analysis

5.3 Identify the root causes:

No	Valid Probable Cause	WHY	WHY	WHY	ACTION / COUNTER MEASURE
1	Associate forgot to fit the socket connection	Number of sockets vary as per variants	Stage associate unable to identify the no. of socket connector	**No reference to identify number of socket connection**	**Provide refernce**
2	Wire trapped in IP	During IP fitment wire gets trapped inside	Due to Critical gap at socket connections area it gets trapped and comes as it to trim 3 line	**No detection for checking available at trim 2 line**	**Check point to be add**
3	Audio socket broken	Due to multiple part handling	Socket gets broken while wiring harness fitment	**No protections available on sockets**	**Provide protection to sockets**
4	Connection poor fit	Stage associate not insert the socket completely.	Stage associate not follow the keypoint	**Stage associate not aware of defect severity.**	**Training to stage associate**

6. Implement the corrective measures

Legends

↑ High ↓ Low

Possible Remedies Identified

S N	Root Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	No reference to identify number of socket connection	Provide refernce	Low ↓	Low ↓	No	Low ↓	Accepted
2	No detection for checking available at trim 2 line	Check point to be add	Low ↓	Low ↓	Yes	Low ↓	Accepted
3	No protections available on sockets	Provide protection to sockets	Low ↓	Low ↓	Yes	Low ↑	Accepted
4	Stage associate not aware of defect severity.	Training to stage associate	Low	Low	Yes	Low	Accepted

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
1	No reference to identify number of socket connection	Reference provided-No of socket connection printed on Sticker.	20.09.19	Ganesh Bendale	Closed

CONT'D..

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
1	No refernce for number of audio socket connection	**Reference provided-No of socket connection printed on Sticker.**	20.09.19	Ganesh Bendale	Closed

HEP CHECKSHEET- HEP ANALYSIS

HUMAN ERROR PREVENTION CHECK LIST Rev - 00

BWT -	6	
Sr.No/Year -	2019	
Date -	19.05.19	
CFT -		
Defect -	AUDIO CONNECTION NOT DONE	

Why	Why	Why	Why	Remark
CONNECTION NOT DONE	ASSO. UNABLE TO IDENTIFY TO AUDIO SOCKET CONNECTOR	NO REFERENCE FOR CONNECTION		REFERNCE TO BE PROVIDED ON STAGE

Principles	Question to be asked to the process	Solution	SPN			
			Effectivness	Cost	Implimentation	SPN
Elimination	Can we eliminate the error-prone process?	Feseabilty to Eliminate connection not done-	3	1	2	6
	Can we eliminate harmful objects?	NO				0
Replacement	Can we automate the process?	NO				0
	Can we provide supporting tools like checklists/gages/samples?	YES	2	3	3	18
Facilitatation	Can we simplify changes/differences in the operations? (standardization, code matching, etc)	YES	2	3	3	18
	Can we distinct changes/differences in the operations? (color coding, unique shape, etc.)	NO				0
	Can We adjust the properties of the operations which suits human beings?	NO				0
Detection	Can we record the motion of the earlier operation and verify it before starting the next operation	YES	2	3	3	18
	Can we restrict the motion in such way that it will notice the abnormility?	NO				0
	Can we verify the results before starting the next operation ? (sensing shape/state/quality of work)	NO				0
Mitigation	Can we make function parelle(Redundant)?(Additional Control)	NO				0
	Can we incorporate some shock absorbing materail/equipment?(Like Fuse)	NO				0
	Can we make unsafe conditions more safer so as to not to produce the defects?(Protection)	NO				0

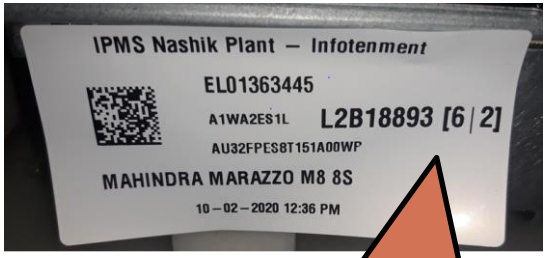
Solution Implimentation

Sr.No	Actions	Resp	Date	Status
1	After Flashing ,Sytem will generate no. of Audio and FATC connection as per model Applicabilty			
2				

ACTION PLAN :

Reference to be provided for Associate,through system generated print indicating No. of socket connection as per model Applicabilty

No of Connection will be printed on Flashing sticker



Referance number printed on success sticker

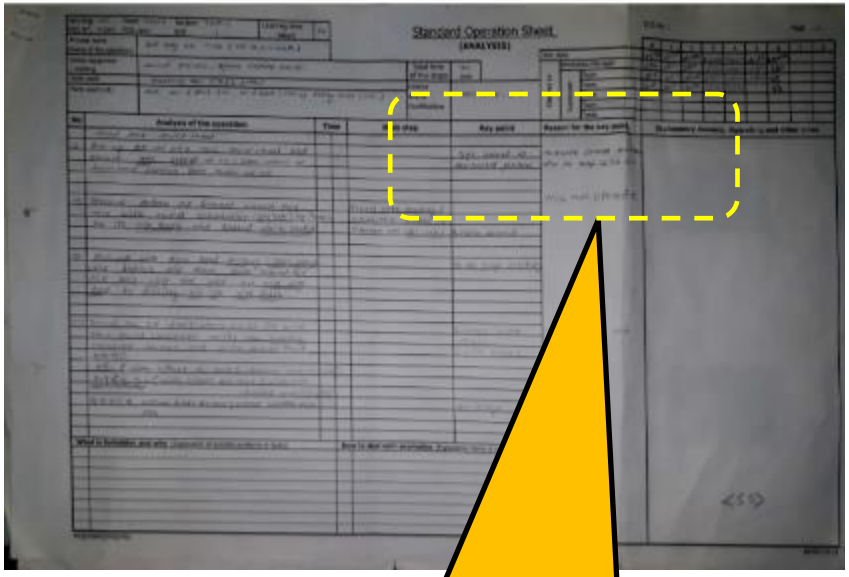
1. After Flashing ,Sytem will generate no. of Audio and FATC connection as per model Applicabilty.
2. Stage associate to confirm the socket connection as displayed on print

6. Implement the corrective measures

6.1 Actions Decided

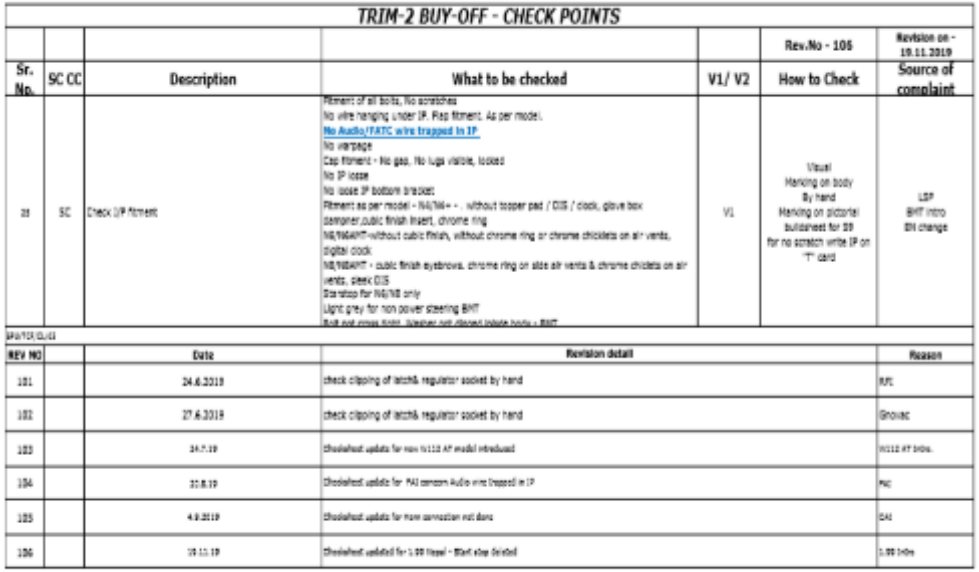
No	Root Cause	Action plan	Actual date	Responsibility	Status
2	No detection for checking available at trim 2 line	1. Keypoint added in SOS. 2. Checkpoint added at BWT3 Buyoff	02.09.19	Ganesh Bendale	Closed

SOS -UPDATION



Key point updated for socket trapped in IP

BUYOFF CHECKSHEET



TRIM-2 BUY-OFF - CHECK POINTS

Rev.No - 106 | Revision on - 19.11.2019

Sr. No.	SC CC	Description	What to be checked	V1/ V2	How to Check	Source of complaint
28	SC	Check I/P fitment	Fitment of all bolts, No scratches No wire hanging under IP. Plug fitment. As per model. No Audio/FATC wire trapped in IP No vampage Cap fitment - No gap, No lugs visible, locked No IP loose No loose IP bottom brackets Fitment as per model - N6/N6++ - without topper pad / DIS / clock, glove box damper,cubic finish insert, chrome ring N8/N8AMT-without cubic finish, without chrome ring or chrome chicklets on air vents, digital clock N8/N8AMT + cubic finish eyebrows, chrome ring on side air vents & chrome chicklets on air vents, sleek DIS Starstop for N6/N8 only Light grey for non power steering BWT Bolt and cross bolt, washer not damaged inside body - BWT	V1	Visual Hanging on body By hand Hanging on pictorial buldsheet for S9 for no scratch write IP on "T" card	LSP BWT intro BN change

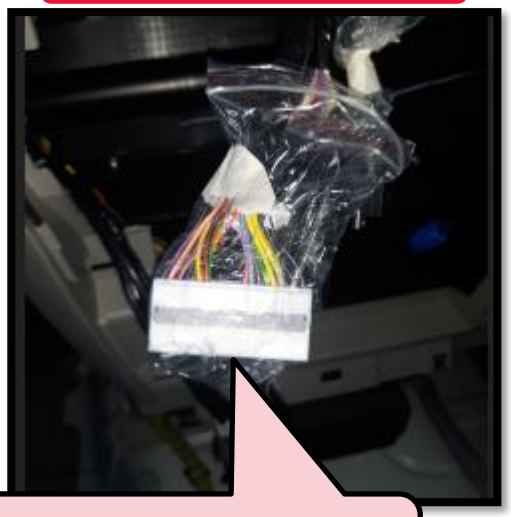
REV NO		Date	Revision detail	Reason
101		24.6.2019	check clipping of latch& regulator socket by hand	RFC
102		27.6.2019	check clipping of latch& regulator socket by hand	Grovac
103		24.7.19	Checksheet update for new N110 AT model introduced	N110 AT Intro.
104		22.8.19	Checksheet update for PAI concern Audio wire trapped in IP	PAI
105		4.9.2019	Checksheet update for more correction not done	DAI
106		19.11.19	Checksheet updated for 1.99 Hippel - Start stop Deleted	1.99 Intro

6. Implement the corrective measures

6.1 Actions Decided

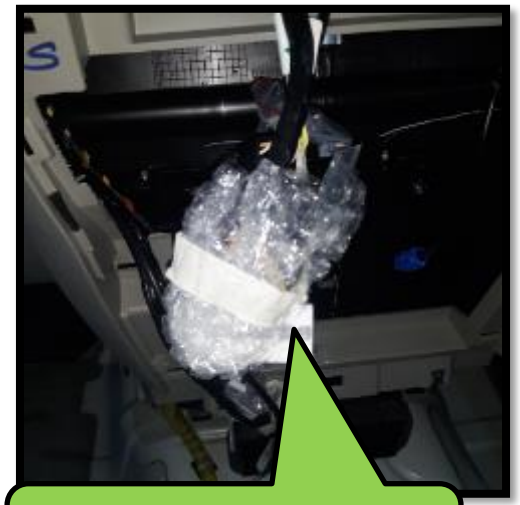
No	Root Cause	Action plan	Actual date	Responsibility	Status
3	No protections available on sockets	Bubble sheet provided on Audio sockets	20.09.19	Deepak Mane	CLOSED

BEFORE



No protection for sockets

AFTER



Bubble sheet provided for Audio sockets

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
3	Stage associate not aware of defect severity-connection poor fit	1. Stage associate to ensure pull-push after socket connection. 2. Keypoint Board displayed at Work-station.	20.09.19	D D Wagh	**CLOSED**

BEFORE

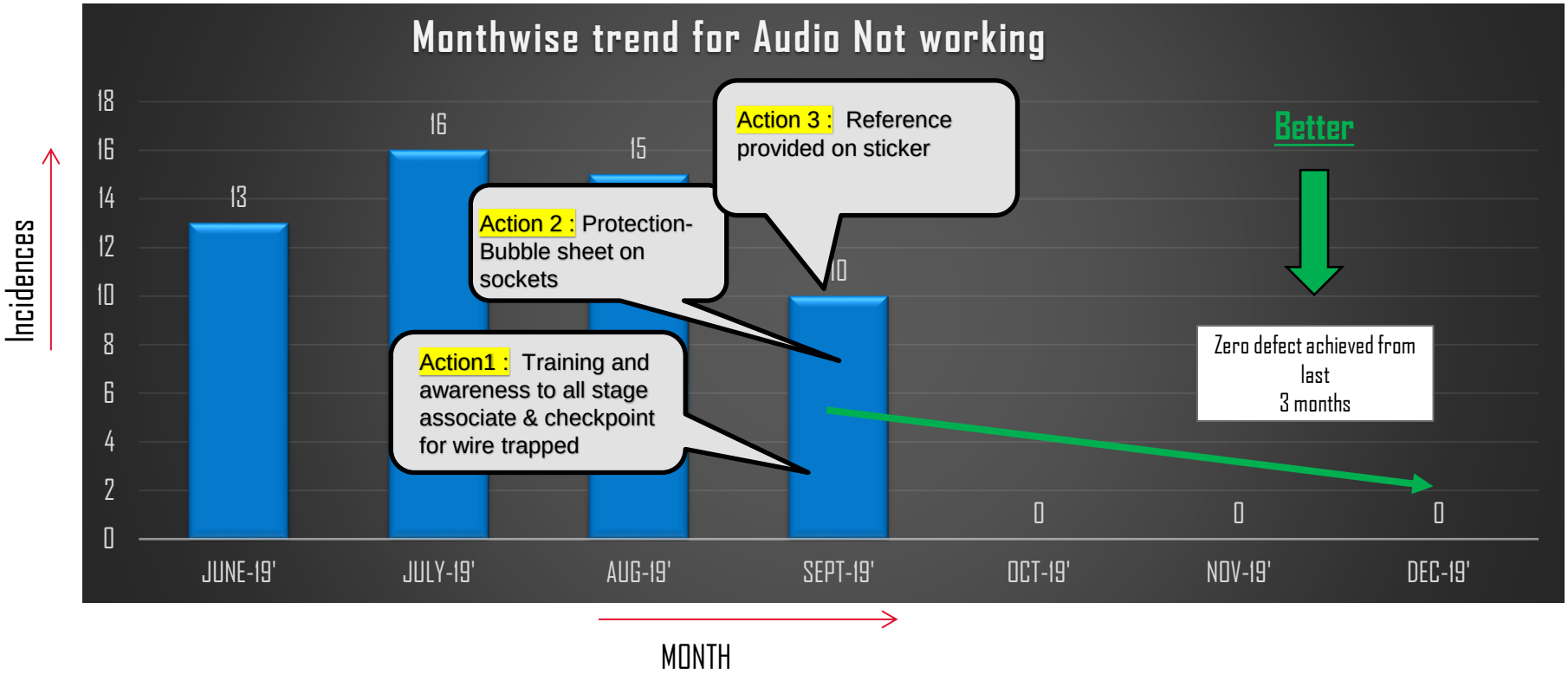
No Keypoint Board Displayed at Work station

AFTER

Keypoint Board Displayed. Emphassis on Process keypoints and Customer effects

7. Check & confirm the results & effects

7.1 Verify Results:



Inference – Concern not reported after implementing the PCA actions.

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Actual date	Responsibility	Status
1	Reference provided-No of socket connection printed on Sticker	20.09.19	Ganesh Bendale	CLOSED
2	SOS correction	02.09.19	Jayesh Jirekar	CLOSED
3	Buyoff checksheet updation	02.09.19	Vikrant Ghule	CLOSED
5	PFMEA updation (Rev No 19)	19.11.19	Ganesh Bendale	CLOSED
6	Process audit check sheet updation	10.09.19	Sachin Bhamare	CLOSED
7	Control plan updation	15.09.19	Vikrant Ghule	CLOSED
9	On job Training given to stage associates for Audio fitment	25.09.19	Dnyaneshwar Wagh	Closed

Check & confirm the results & effects (Cont……)

7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) Elimination of concern Audio not working
2) Rework cost save upto Rs 19103.2 / Year
3) FRC improved and FBO loss improved

In tangible Benefits

1) Improved Customer Satisfaction.
2) Operator fatigue reduced , moral improved

Cost Approval mail from Industrial engineering

Mon 2/10/2020 5:57 PM
DR DHANOKAR RAJESH
RE: MYB project cost sheet approval
To WAGH DNYANESHWAR
Cc BENDALE GANESH

Repair /Restoration Cost estimation		
Description	Units	
Rework time (Approx)	Min	30
Avg Concerns/Day	Nos	0.5
Total Required Rework time per day	Min	15
Effective working time	Min	465
Operator required /day	nos	0.03
Approx Operator cost/ day	Rs	1974
Operator cost /concern	Rs	63.68
Avg no. of working days	days	25
Operator cost /month	Rs	1591.9
Operator Cost/year	**Rs**	**19103.2**

From: WAGH DNYANESHWAR
Sent: Monday, February 10, 2020 5:29 PM
To: DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
Cc: BENDALE GANESH <BENDALE.GANESH@mahindra.com>
Subject: MYB project cost sheet approval

Type here to search

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

S N	Document	Target date	Responsibility	Status
1	Reference provided-No of socket connection printed on Sticker	20.09.19	Ganesh Bendale	CLOSED
2	SOS correction	02.09.19	Jayesh Jirekar	CLOSED
3	Buyoff checksheet updation	02.09.19	Vikrant Ghule	CLOSED
4	PFMEA updation (Rev No 19)	19.11.19	Ganesh Bendale	CLOSED

8.2 Training:

SN	Training Identified	Target date	Responsibility	Status
1.	On job Training given to stage associates for Audio fitment and connections.	25.09.19	Ganesh Bendale	Closed

8.3 Additional Audit & check :

SN	Document	Target date	Responsibility	Status
1.	Daily Process audit sheet updated	01.09.19	Ganesh Bendale	Closed
2	As per ILU plan, Stage Associate audit done, for adherence to key-point-No NC found	15.10.19	Ganesh Bendale	Closed

8. Standardization

8.1 Updation of documents :-

PFMEA Updation

2	Part No/s.:		Part Name/ Process: Audio system.	Mahindra & Mahindra Ltd.										$ $ SHELKE				
3	Latest Change Letter:NA		Opta No.: 0970	Process Potenial Failure Mode & Effect Analysis.										Mandar Joshi				
4	PU/ Shop :Scorpio TCF		Opta Despta: Audio Fitment.	Reviewed by :$ $SHELKE							Rev. No :	18	Doc No.: PFMEA/SPU/SCORPIO/TRIM/0970	Vikrant Ghule				
5	Module/ Cell :Trim-2		M/C Name & No.:_NA	Approved by : Mandar Joshi							Rev. Date :	5.08.15	FMEA Date (Orig) :05/12/05	Vikrant Morade				
6	Process Step	Requirements	Potential Failure Mode	Potential Effect(s) of Failures	sev	class	Potential Cause(s)/Mechanism(s) of Failures	Current Process Controls Prevention	Occur	Current Process Controls Detection	Detect	RPN	Recommended Action(s)	Responsibility & Target Completion Date	Action Results			
7	Function														Action Taken	Sev	Occ	Det	RPN
8		Collection of Audio as per Model	Wrong collection of Audio (Not as per model)	**Mfg & Assy:** rework. **End User: Features mismatch**	6		wrong sequencing by SCM, operator error by selecting audio by disturbing the sequence.	variant coding will not start if wrong audio selected.	4	Self certification by stage associte	2	48	Inbult Pokayoke for Audio Flashing	**Sayali More**				
9		Connection of antenna cable & wire loom ,USB socket socket in to audio. **(FAI Concern Aug 19)**	Socket not inserted fully	**Mfg & Assy:** Rework. **End User:** Audio inoperative (6)	6	SC	Operator doesn't press sockets fully - Human error	operator awareness & traning. Click sound of socket	3	Checking at trim3, final Buyoff & CAI.	7	126	**Associte to ensure number of socket connection as per printed on sticker**	**Ganesh Bendale 20.09.19**		6	2	5	60
10	Audio Fitment .	Correct connections Connection of MIC cable with Audio	Socket not inserted fully	**Mfg & Assy:** rework **End User:** MIC inoperative	6	SC	Operator doesn't the press sockets fully in Audio Operator failed to sense complet locking since there is no positive locking of the connector.	operator awareness & education.	4	Checking at trim3, final Buyoff & CAI.	7	168	**positive locking to be provided for MIC cable with audio. Else clipping to be provided for MIC cable on audio, so that there will not be direct pull on the socket connection**	**Prasad Bhambore**		5	4	6	120

8. Standardization

8.1 Updation of documents :-

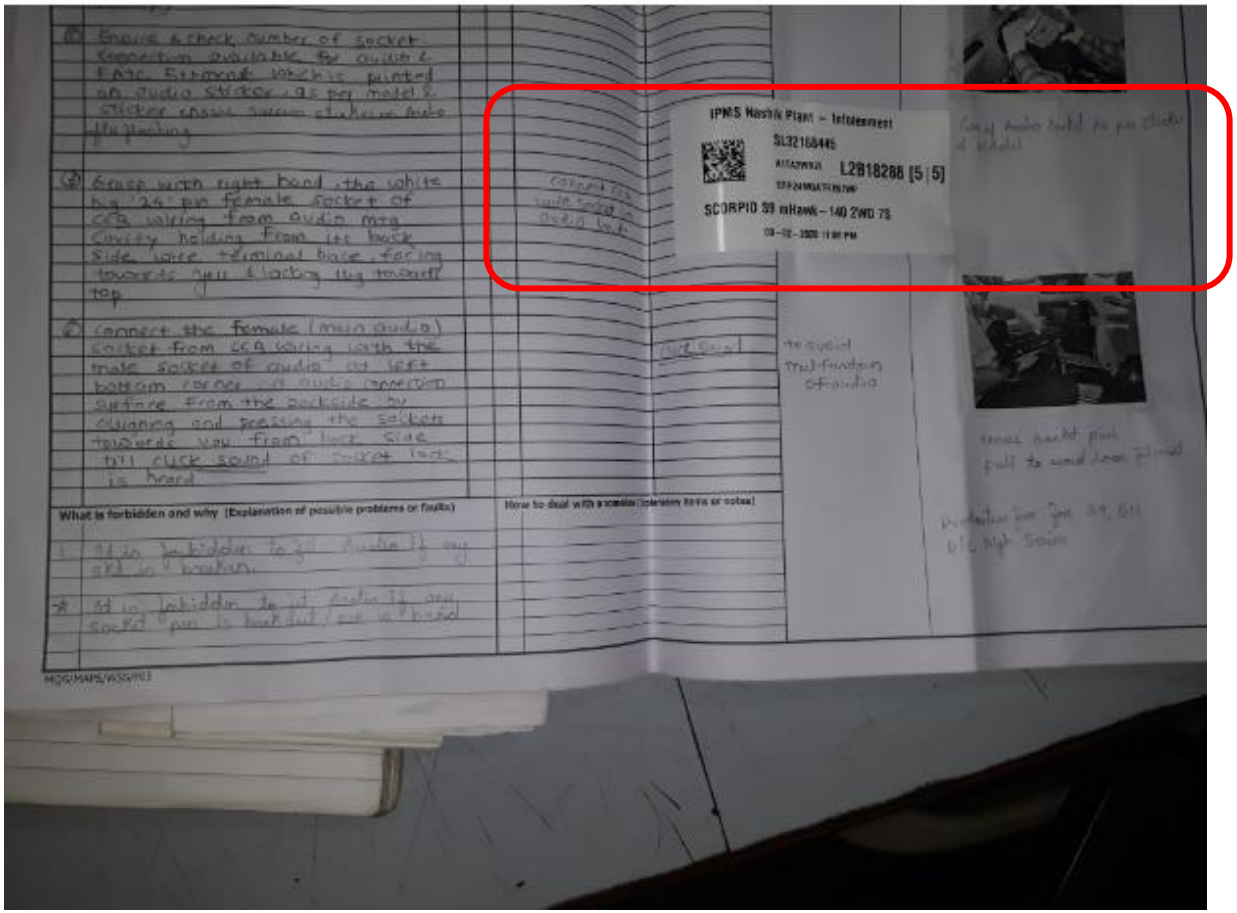
CONTROL PLAN Updation

1													
2	Plant : Nashik		Model :Scorpio& BMT	Mahindra & Mahindra Ltd.									
3	Part No/s.: -0101AG0010NTC2T5		Part Name/ Process: Trim Line	Control Plan									Vikrant Ghule
4	Latest Change Letter:-		Doc No.:	Prototype : ☐			Prelaunch: ☐		Production: ☐		Critical Char: <CC>		Sakharam Shelke
5	Aggregate No.:		Aggregate Part Descrptn :	Module Name : Scorpio Assy.			Reviewed by :Vikrant Ghule		Rev. No : 115		Significant Char : <SC>		Pravin Patil
6	PU/ Shop :Scorpio/Xylo TCF		Next Aggregate No.:	Cell Name :Trim 3			Approved by : Mandar joshi		Rev. Date : 28.09.18				C
7	Opn No.	Operation Description	Machine, Device, Jig, Tools Description	Characteristics			Sp. Chara. Class	Method					
8				No.	Product	Process		Product/ Process Specification.	Evaluation Measurement Tech.	Sample		Control Met	
9										Size	Freq.	Record	
607	0970	Audio Fitment		1	Fitment as per model	Sequential feeding		LCD for S11, W109 high, body number to be matched	Visual	100%	Daily	Traveller card	A:
608				2	CD insertion in audio	Removal of C.D. mechanism locking screw from audio		As per SOS	Visual	100%	Daily	Traveller card	A:
609				3	Socket connection	Press socket fully for FATC & Infotainment/Audio		Click sound USB/AUX - Push pull	Visual	100%	Daily	Traveller card	
			STCF/TOOLS/TRIM3/12_Rev00	4	No damage to front panel	Use of fixture for fitment		Holding with fixture while fitment	Visual	100%	Daily	Traveller card	A:

8. Standardization

8.1 Updation of documents :-

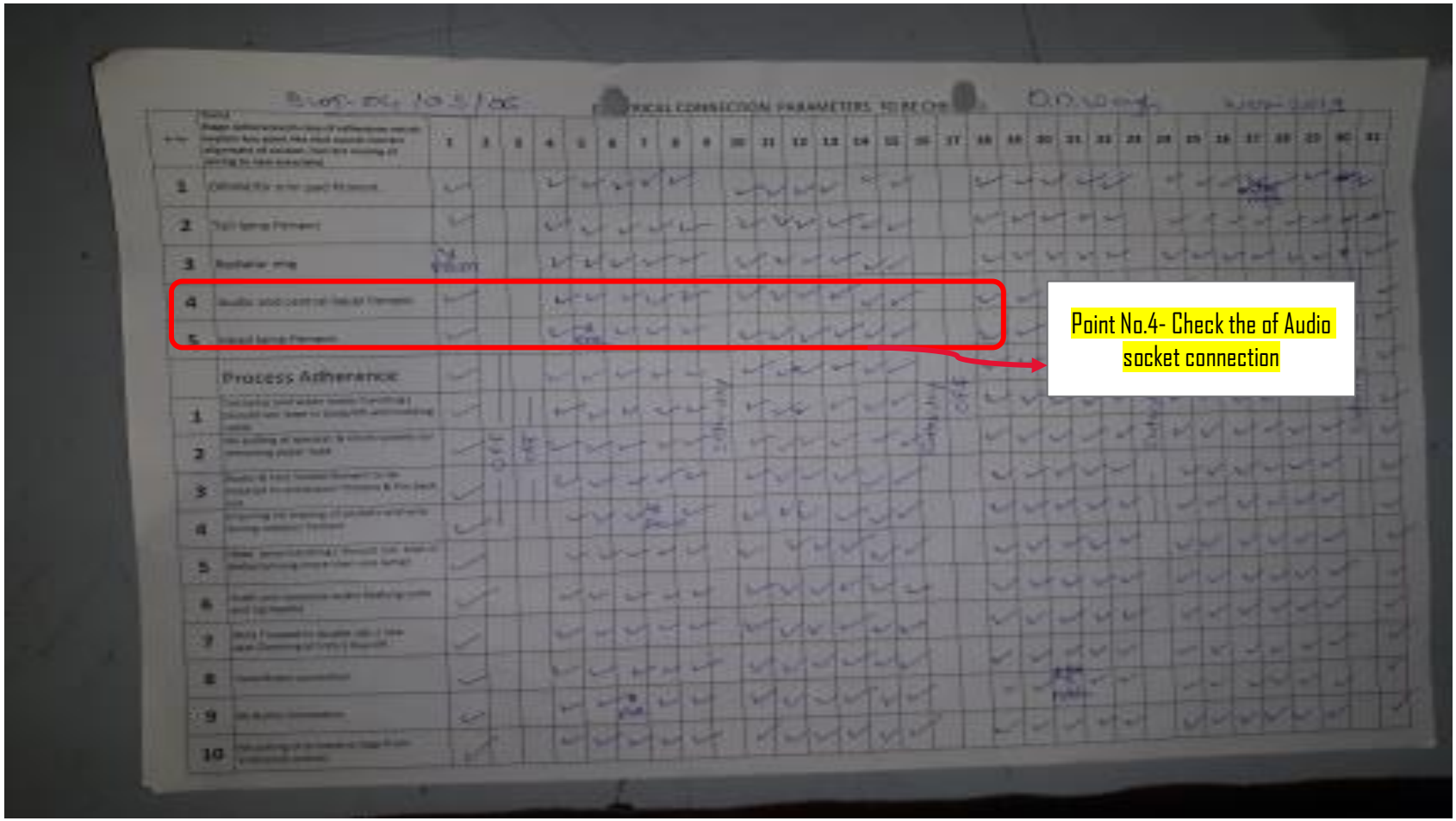
SOS Updation



8. Standardization

8.3 Additional Audit & check

Daily Process audit check sheet updation :-



9. Summarize & Plan future actions

9.1 Conclusion:

The concern of Audio not working due to socket connection not done eliminated,by implementation/provision of reference to stage associate to identify no. of socket connection

9.2 Horizontal Deployment:

All Scorpio/Export/Marazzo model where Audio fitment is applicable.

9.3 Lessons Learnt:

Things Gone Wrong : Facility not provided to stage associate for easy identification of Number of socket connection as per variants.

Things Gone Right : Understood the importance of HEP analysis and its mandatory consideration in process.

Thank you

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